

AddisonCoriSpring2026

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Exploratory Data Analysis using data from petri dishes 1 and 2 respectively

This document explores the data set which seeks to determine whether there is a significant difference in effect between fluoride tooth paste and nano-hydroxide toothpaste. The experiment considered three different concentration levels of fluoride (high - 1 : 10, medium - 1 : 100, low - 1 : 1000) respectively. There were also two other treatments, BHI and BHI with Streptococcus, where the latter served as control. Treatments were randomly assigned to cultures in each column of a petri dish. For each treatment in each column, 4 rows also received sucrose. In this data, trt 1 to 8 represents BHI, BHI+Strep, Nano-Oxide at high concentration, Nano-Oxide at medium concentration, Nano-Oxide at low concentration, Fluoride at high concentration, Fluoride at medium concentration, and Fluoride at low concentration respectively. The response variable y measures the amount of reduction in bacteria by each treatment. Each treatment (fluoride or nano-oxide) together with a given concentration was considered as a separate treatment due to the nature of the experiment. The experiment is analyzed as a *completely randomized design with factorial arrangement of treatments* where the factors are Treatment and X, each with 8 and 2 levels respectively.

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##   recode
```

The section below converts the data from wide to a long format and treatment factors, X (sucrose, no sucrose) and Treatment (trt1 to 9) are treated as such. The first 6 rows of the data is shown below.

```
# Reading in the first data set
petri1 <- read.csv("Addison.csv")

# changing from wide to long format

petri1_long <- petri1 %>%
  pivot_longer(
    cols = starts_with("trt"),
    names_to = "Treatment",
```

```

    values_to = "Y"
  )
petri1_long$Treatment <- as.factor(petri1_long$Treatment)
petri1_long$X <- as.factor(petri1_long$X)
head(petri1_long)

```

```

## # A tibble: 6 x 3
##   X      Treatment      Y
##   <fct>    <fct>    <dbl>
## 1 no sucrose trt1      0.433
## 2 no sucrose trt2      1.09
## 3 no sucrose trt3      2.25
## 4 no sucrose trt4      0.42
## 5 no sucrose trt5      2.01
## 6 no sucrose trt7      2.63

```

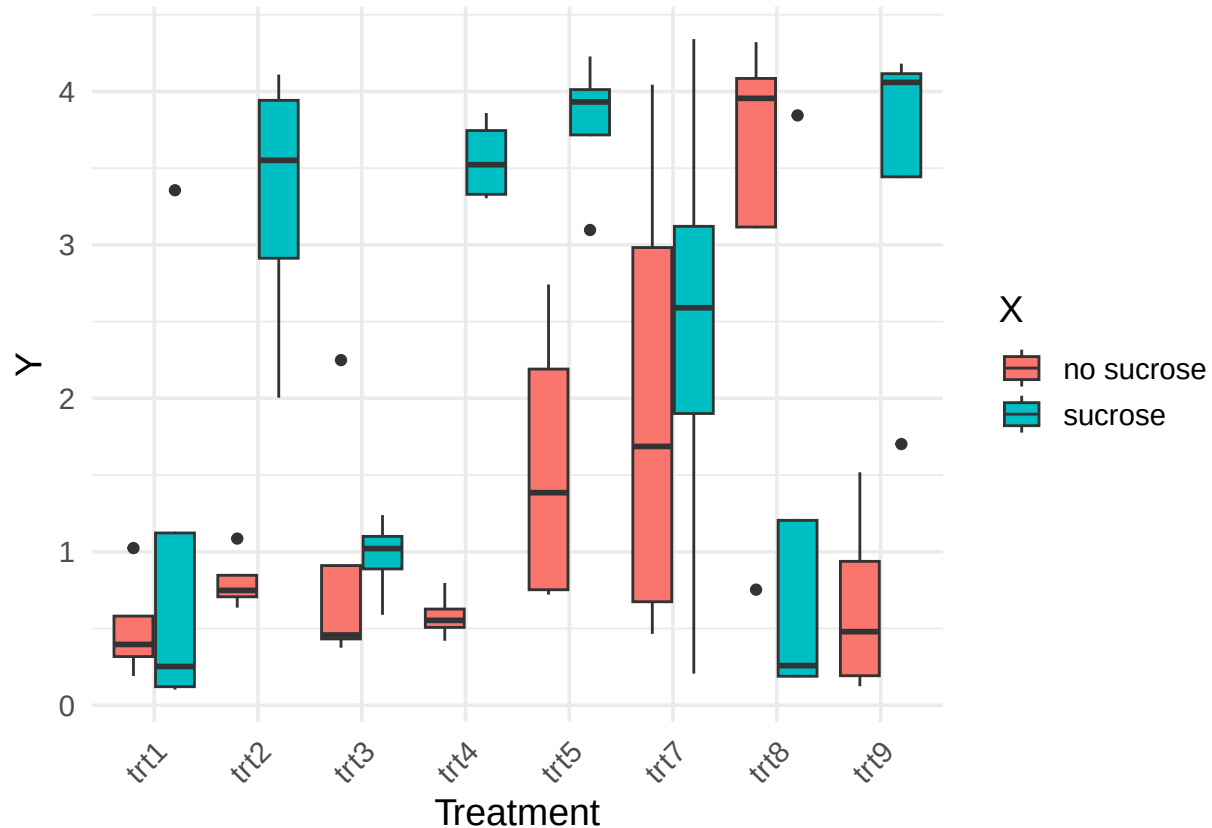
Next, we show the boxplot of each Treatment and X. The boxplots in orange are treatments with no sucrose and those in green are treatments with sucrose. From the boxplots, treatments without sucrose generally seems to have lower mean than their counterparts except for treatment 8 (Fluoride with medium concentration) where treatment with sucrose had a lower mean when data from petri dish 1 was used. Nevertheless, there is high variability in the data. This is shown by the size of the boxplots for both treatments with sucrose and without sucrose.

```

# Boxplot

ggplot(petri1_long, aes(x = Treatment, y = Y, fill = X)) +
  geom_boxplot(position = position_dodge(0.8)) +
  labs(x = "Treatment", y = "Y", fill = "X") +
  theme_minimal(base_size = 14) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

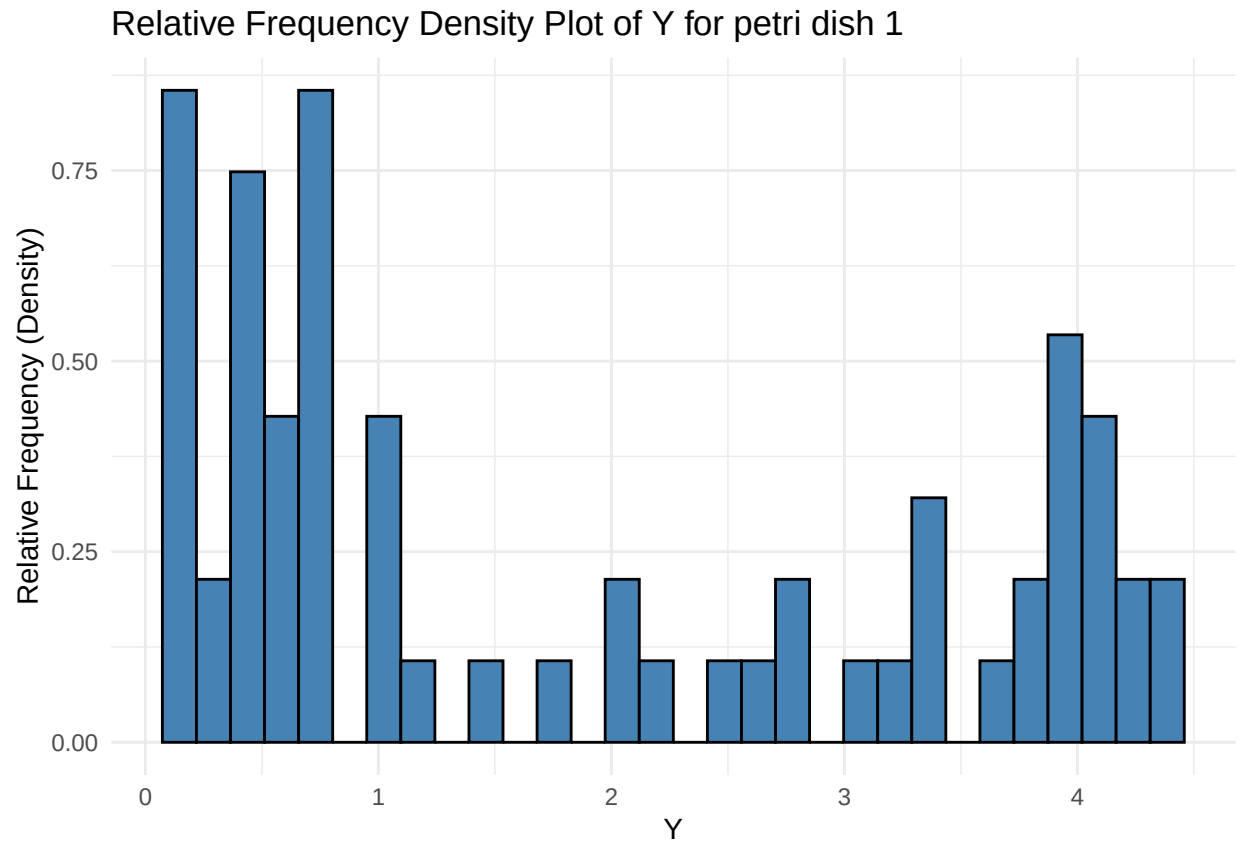


Distribution of y for data set in petri dish 1

We also show the distribution of the response through the relative frequency density plot. The plot shows a bi-modal shape (two peaks) for this data. This means that the response variable *does not* follow a normal distribution and thus, we need to account for the distribution in our analysis.

```
# Distribution of response
ggplot(petri1_long, aes(x = Y)) +
  geom_histogram(aes(y = ..density..),
                 bins = 30,
                 fill = "steelblue",
                 color = "black") +

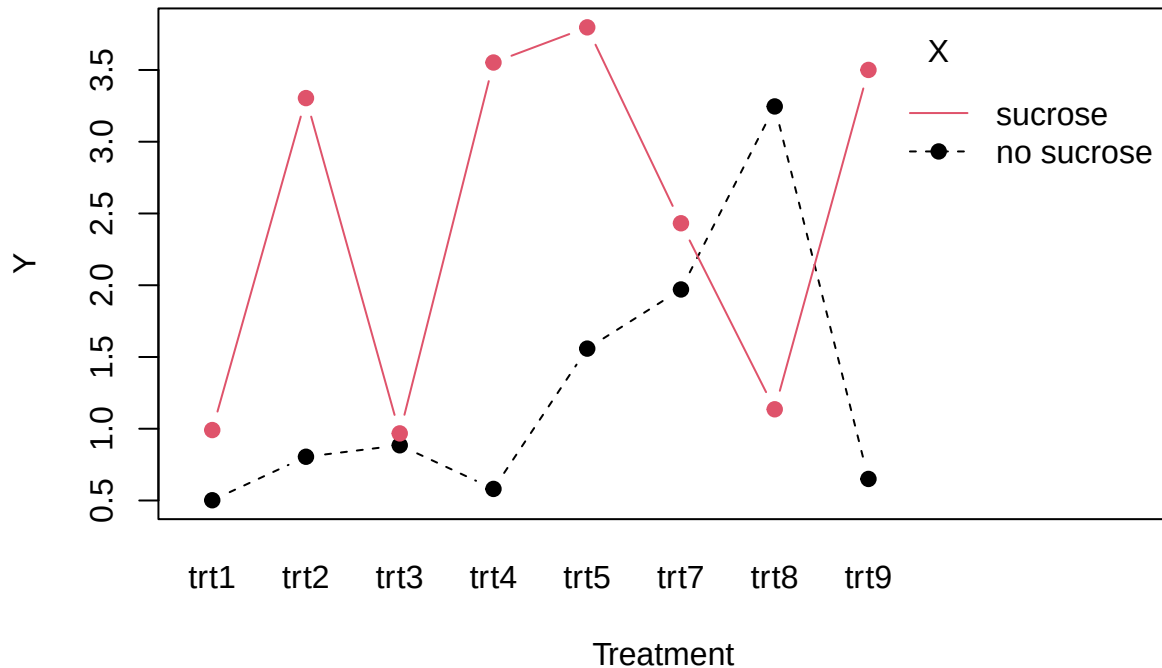
  labs(
    x = "Y",
    y = "Relative Frequency (Density)",
    title = "Relative Frequency Density Plot of Y for petri dish 1"
  ) +
  theme_minimal()
```



Lastly, we show an interaction plot using data from petri dish 1. There is an interaction between the two factors. What this means is that, the effect of a particular treatment depend on whether sucrose was added or not. We still need to analyze the data to confirm whether the interaction is statistically significant.

```
# Interaction plot
with(petri1_long, interaction.plot(
  x.factor = Treatment,
  trace.factor = X,
  response = Y,
  type = "b",
  col = 1:5,
  pch = 19,
  xlab = "Treatment",
  ylab = "Y",
  trace.label = "X",
  main = "Interaction plot of Treatment and X"
))
```

Interaction plot of Treatment and X



Model fit for data on petri dish 1

We fit a generalized linear model for data on petri dish 1 using the hurdle model due to the bi-modal nature of the data. Fixed effects of X, Treatment, and their interaction were tested using Chi-square. Estimated marginal means were obtained on the response scale with 95% confidence intervals. The Analysis of Deviance Table shows that there is a significant Treatment by X (sucrose or no sucrose) interaction. Therefore, finding the main effect of treatments is *not* appropriate since the effect of each treatment depends on whether sucrose was used or not. Nevertheless, the simple effects, that is, the effect of each treatment at a given level of sucrose or no sucrose can be obtained. Thus, for example we can compare the effect of fluorides that had sucrose to nano-oxides that had sucrose and vice versa. But we can not compare fluorides with sucrose with nano-oxides without sucrose. The Chi-square test was used to determine if there is any significant effect from any or all of the treatment factors.

```
# Model Fit for Petri dish 2 data
```

```
hurdle_fit <- glmmTMB(
  Y ~ X * Treatment,
  ziformula = ~ X * Treatment,
  family = Gamma(link = "log"),
  data = petri1_long
)
```

```
# Get anova table
```

```
Anova(
  hurdle_fit,
```

```

type = 3
)

## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: Y
##           Chisq Df Pr(>Chisq)
## (Intercept)  3.8986  1  0.0483274 *
## X           1.8971  1  0.1684030
## Treatment   25.0048  7  0.0007573 ***
## X:Treatment  26.3094  7  0.0004434 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# Get least square means
lsm0 <- emmeans(hurdle_fit, ~ X * Treatment, type = "response")
lsm0

## X           Treatment response      SE df asymp.LCL asymp.UCL
## no sucrose trt1           0.502 0.175 Inf      0.253      0.995
## sucrose   trt1           0.991 0.346 Inf      0.500      1.965
## no sucrose trt2           0.805 0.281 Inf      0.406      1.596
## sucrose   trt2           3.304 1.154 Inf      1.666      6.552
## no sucrose trt3           0.885 0.309 Inf      0.446      1.754
## sucrose   trt3           0.968 0.338 Inf      0.488      1.918
## no sucrose trt4           0.581 0.203 Inf      0.293      1.152
## sucrose   trt4           3.552 1.241 Inf      1.791      7.043
## no sucrose trt5           1.558 0.544 Inf      0.786      3.090
## sucrose   trt5           3.797 1.326 Inf      1.915      7.529
## no sucrose trt7           1.970 0.688 Inf      0.994      3.907
## sucrose   trt7           2.432 0.849 Inf      1.226      4.822
## no sucrose trt8           3.246 1.134 Inf      1.637      6.437
## sucrose   trt8           1.135 0.397 Inf      0.573      2.252
## no sucrose trt9           0.650 0.227 Inf      0.328      1.289
## sucrose   trt9           3.500 1.223 Inf      1.765      6.941
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale

```

Contrast Statements

Since there was a significant interaction effect, we can only look at the simple and interaction effects. This is because looking at the main effects does not give any useful information. The table below shows specific contrasts that the client was interested in. Since the client is interested in treatments with sucrose, we show the Differences of X*Treatment Least Squares Means in the table below. In particular, we are interested in comparison between treatments (3 vs 7), (4 vs 8) and (5 vs 9). From the table, we see that the p-values for treatments (3 vs 7), (4 vs 8) and (5 vs 9) are 0.5746, 0.2886, and 1.0000 respectively. This means that for data on petri dish 1, there are no significant differences between these treatments.

```

# EMMs only for sucrose
lsm0 <- emmeans(
  hurdle_fit,
  ~ Treatment,
  at = list(X = "sucrose"),
  type = "response"
)

```

```
)
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
# Pairwise treatment differences (with p-values)
```

```
pairs(lsm0)
```

```
## contrast      ratio      SE df null z.ratio p.value
## trt1 / trt2 0.300 0.148 Inf    1 -2.438 0.2228
## trt1 / trt3 1.024 0.506 Inf    1  0.048 1.0000
## trt1 / trt4 0.279 0.138 Inf    1 -2.585 0.1613
## trt1 / trt5 0.261 0.129 Inf    1 -2.720 0.1165
## trt1 / trt7 0.407 0.201 Inf    1 -1.818 0.6079
## trt1 / trt8 0.873 0.431 Inf    1 -0.276 1.0000
## trt1 / trt9 0.283 0.140 Inf    1 -2.555 0.1727
## trt2 / trt3 3.415 1.687 Inf    1  2.486 0.2011
## trt2 / trt4 0.930 0.459 Inf    1 -0.147 1.0000
## trt2 / trt5 0.870 0.430 Inf    1 -0.282 1.0000
## trt2 / trt7 1.359 0.671 Inf    1  0.621 0.9986
## trt2 / trt8 2.910 1.437 Inf    1  2.162 0.3749
## trt2 / trt9 0.944 0.466 Inf    1 -0.117 1.0000
## trt3 / trt4 0.272 0.135 Inf    1 -2.633 0.1441
## trt3 / trt5 0.255 0.126 Inf    1 -2.768 0.1031
## trt3 / trt7 0.398 0.197 Inf    1 -1.866 0.5746
## trt3 / trt8 0.852 0.421 Inf    1 -0.324 1.0000
## trt3 / trt9 0.276 0.137 Inf    1 -2.603 0.1546
## trt4 / trt5 0.935 0.462 Inf    1 -0.135 1.0000
## trt4 / trt7 1.461 0.722 Inf    1  0.767 0.9947
## trt4 / trt8 3.128 1.545 Inf    1  2.309 0.2886
## trt4 / trt9 1.015 0.501 Inf    1  0.030 1.0000
## trt5 / trt7 1.561 0.771 Inf    1  0.902 0.9860
## trt5 / trt8 3.344 1.652 Inf    1  2.444 0.2202
## trt5 / trt9 1.085 0.536 Inf    1  0.165 1.0000
## trt7 / trt8 2.142 1.058 Inf    1  1.542 0.7848
## trt7 / trt9 0.695 0.343 Inf    1 -0.737 0.9959
## trt8 / trt9 0.324 0.160 Inf    1 -2.279 0.3051
##
## P value adjustment: tukey method for comparing a family of 8 estimates
## Tests are performed on the log scale
```

Exploratory Data Analysis for petri dish 2

We use a similar process to transform our data from wide to long format. Again, the first six rows are show below.

```
# Reading in the data
petri2 <- read.csv("petri2.csv")

# changing from wide to long format

petri2_long <- petri2 %>%
  pivot_longer(
    cols = starts_with("trt"),
    names_to = "Treatment",
    values_to = "Y"
```

```
)

# Treating X and Treatment as factors
petri2_long$Treatment <- as.factor(petri2_long$Treatment)
petri2_long$X <- as.factor(petri2_long$X)
head(petri2_long)

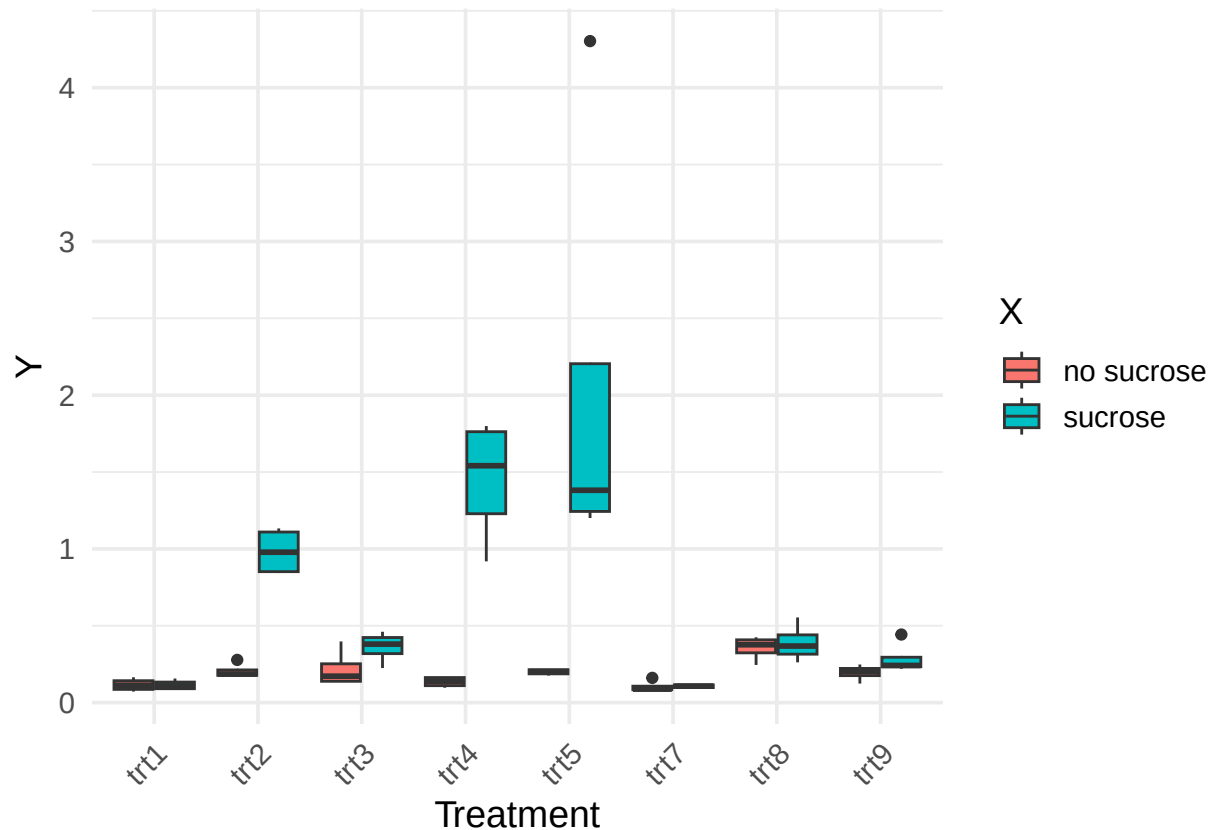
## # A tibble: 6 x 3
##   X      Treatment      Y
##   <fct>    <fct>    <dbl>
## 1 no sucrose trt1      0.091
## 2 no sucrose trt2      0.278
## 3 no sucrose trt3      0.14
## 4 no sucrose trt4      0.097
## 5 no sucrose trt5      0.217
## 6 no sucrose trt7      0.077

#write.csv(petri2_long, "petri2_long.csv")
```

We now consider data from petri dish 2 and show the boxplots of each Treatment and X. Again, boxplots in orange are treatments with no sucrose and those in green are treatments with sucrose. From the boxplots, treatments without sucrose generally seems to have lower mean than their counterparts except for treatment 8 (Fluoride with medium concentration) where treatment with sucrose had about the same mean as treatment without sucrose. Nevertheless, there is less variability with treatments without sucrose compared to treatments with sucrose in this data. This is shown by the size of the boxplots for both treatments with sucrose and without sucrose respectively.

```
# Boxplot

ggplot(petri2_long, aes(x = Treatment, y = Y, fill = X)) +
  geom_boxplot(position = position_dodge(0.8)) +
  labs(x = "Treatment", y = "Y", fill = "X") +
  theme_minimal(base_size = 14) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

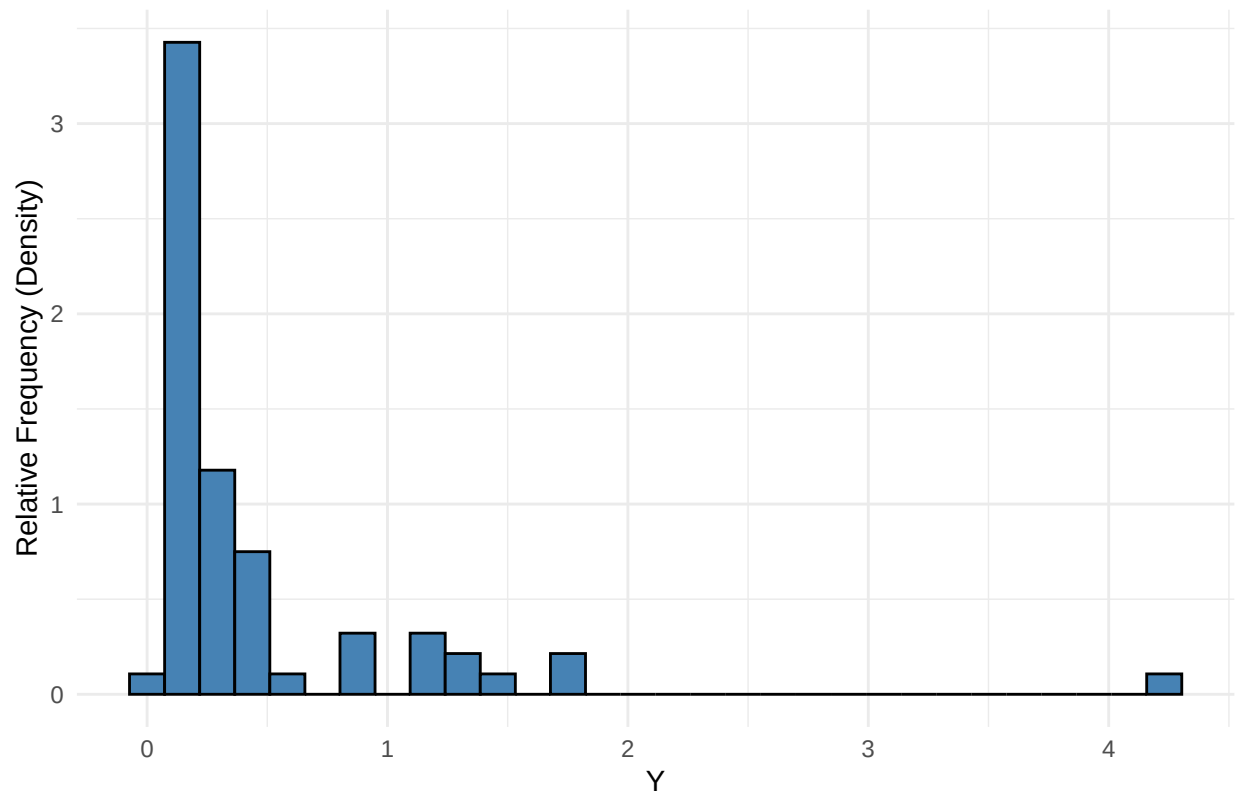


Distribution of data in petri dish 2

For data in petri dish 2, we see a strong right skewness. This means that the assumption of normality again fails and thus, the appropriate distribution should be used. We consider the gamma distribution as an appropriate distribution for this data.

```
# Distribution of response
ggplot(petri2_long, aes(x = Y)) +
  geom_histogram(aes(y = ..density..),
    bins = 30,
    fill = "steelblue",
    color = "black") +
  labs(
    x = "Y",
    y = "Relative Frequency (Density)",
    title = "Relative Frequency Density Plot of Y for petri dish 1"
  ) +
  theme_minimal()
```

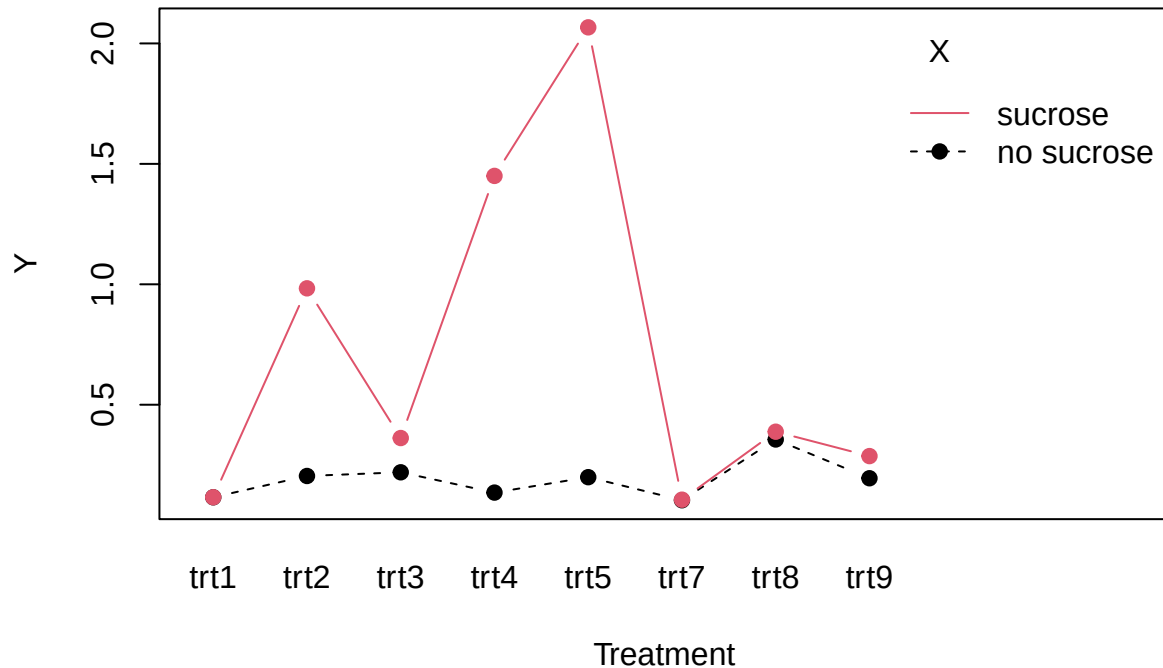
Relative Frequency Density Plot of Y for petri dish 1



Lastly for a factorial design, we again check for any interaction effect. We see that the lines are not parallel indicating an interaction between treatments and whether a sucrose was added or not. In other words, the effect of each of the treatments depends on whether sucrose was added or not.

```
# Interaction plot
with(petri2_long, interaction.plot(
  x.factor = Treatment,
  trace.factor = X,
  response = Y,
  type = "b",
  col = 1:5,
  pch = 19,
  xlab = "Treatment",
  ylab = "Y",
  trace.label = "X",
  main = "Interaction plot of Treatment and X"
))
```

Interaction plot of Treatment and X



Model Fit for data on petri dish 2

We fit a generalized linear model for data in petri dish 2 using the gamma distribution which uses the log as its link function. Fixed effects of X, Treatment, and their interaction were tested using Type III F-tests. Estimated marginal means were obtained on the response scale with 95% confidence intervals. The anova table is shown below. The anova table shows a significant interaction effect between Treatment and X (no sucrose or sucrose). Since there is a significant interaction effect, it is not advisable to find the main effect of treatment since the effect of a given treatment or toothpaste depends on whether sucrose was added. Therefore, we will only consider the simple effects - effect of each treatment at a given level of sucrose or no sucrose.

Also, this means that we can compare for example, fluorides with sucrose to nano-oxide with sucrose but we cannot compare fluoride with sucrose to nano-oxides without sucrose.

Model Fit for Petri dish 2 data

```
glm_fit <- glm(  
  Y ~ X * Treatment,  
  data = petri2_long,  
  family = Gamma(link = "log")  
)
```

Get anova table

```
Anova(  
  glm_fit,  
  type = 3,  
  test.statistic = "F"
```

```
)

## Analysis of Deviance Table (Type III tests)
##
## Response: Y
## Error estimate based on Pearson residuals
##
##          Sum Sq Df F values    Pr(>F)
## X          0.0000  1    0.0001 0.9929332
## Treatment    4.5508  7    5.5356 0.0001037 ***
## X:Treatment 13.8767  7   16.8794 5.128e-11 ***
## Residuals    5.6373 48
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# Get least square means
lsm <- emmeans(glm_fit, ~ X * Treatment, type = "response")
lsm
```

```
## X          Treatment response      SE df lower.CL upper.CL
## no sucrose trt1          0.116 0.0198 48   0.0820   0.163
## sucrose    trt1          0.116 0.0199 48   0.0822   0.164
## no sucrose trt2          0.204 0.0350 48   0.1447   0.288
## sucrose    trt2          0.983 0.1685 48   0.6967   1.388
## no sucrose trt3          0.220 0.0376 48   0.1555   0.310
## sucrose    trt3          0.362 0.0620 48   0.2563   0.511
## no sucrose trt4          0.135 0.0232 48   0.0958   0.191
## sucrose    trt4          1.450 0.2485 48   1.0274   2.046
## no sucrose trt5          0.199 0.0341 48   0.1412   0.281
## sucrose    trt5          2.067 0.3541 48   1.4644   2.917
## no sucrose trt7          0.103 0.0177 48   0.0733   0.146
## sucrose    trt7          0.106 0.0181 48   0.0749   0.149
## no sucrose trt8          0.356 0.0610 48   0.2521   0.502
## sucrose    trt8          0.388 0.0665 48   0.2749   0.548
## no sucrose trt9          0.195 0.0334 48   0.1380   0.275
## sucrose    trt9          0.287 0.0491 48   0.2032   0.405
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

Contrasts comparing treatments with sucrose

In the second data, the client is also interested in treatments with sucrose, and thus, we show the Differences of X*Treatment Least Squares Means in the table below. In particular, we are interested in comparison between treatments (3 vs 7), (4 vs 8) and (5 vs 9). Looking at the table and the relevant contrasts, we see that there is a significant difference in treatment effect between treatments 3 and 7, 4 and 8, 5 and 9. The p-values obtained using the student t-test are 0.0002, and 0.0001 respectively.

Also, if the researcher wanted to know how the level of concentration affected these treatments, we can also look at the contrasts for treatments (3 vs 4), (3 vs 5), (4 vs 5), (7 vs 8), (7 vs 9) and so on.

```
# EMMs only for sucrose
lsm_sucrose <- emmeans(
  glm_fit,
  ~ Treatment,
  at = list(X = "sucrose"),
```

```

type = "response"
)

```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
# Pairwise treatment differences (with p-values)
```

```
pairs(lsm_sucrose)
```

##	contrast	ratio	SE	df	null	t.ratio	p.value
##	trt1 / trt2	0.1180	0.0286	48	1	-8.820	<.0001
##	trt1 / trt3	0.3207	0.0777	48	1	-4.694	0.0006
##	trt1 / trt4	0.0800	0.0194	48	1	-10.423	<.0001
##	trt1 / trt5	0.0561	0.0136	48	1	-11.885	<.0001
##	trt1 / trt7	1.0969	0.2658	48	1	0.382	0.9999
##	trt1 / trt8	0.2990	0.0724	48	1	-4.983	0.0002
##	trt1 / trt9	0.4045	0.0980	48	1	-3.735	0.0108
##	trt2 / trt3	2.7180	0.6587	48	1	4.126	0.0034
##	trt2 / trt4	0.6781	0.1643	48	1	-1.603	0.7461
##	trt2 / trt5	0.4757	0.1153	48	1	-3.066	0.0644
##	trt2 / trt7	9.2979	2.2531	48	1	9.202	<.0001
##	trt2 / trt8	2.5341	0.6141	48	1	3.837	0.0081
##	trt2 / trt9	3.4289	0.8309	48	1	5.085	0.0002
##	trt3 / trt4	0.2495	0.0605	48	1	-5.729	<.0001
##	trt3 / trt5	0.1750	0.0424	48	1	-7.192	<.0001
##	trt3 / trt7	3.4208	0.8289	48	1	5.075	0.0002
##	trt3 / trt8	0.9323	0.2259	48	1	-0.289	1.0000
##	trt3 / trt9	1.2616	0.3057	48	1	0.959	0.9780
##	trt4 / trt5	0.7016	0.1700	48	1	-1.463	0.8229
##	trt4 / trt7	13.7116	3.3227	48	1	10.805	<.0001
##	trt4 / trt8	3.7371	0.9056	48	1	5.440	<.0001
##	trt4 / trt9	5.0567	1.2254	48	1	6.688	<.0001
##	trt5 / trt7	19.5437	4.7360	48	1	12.267	<.0001
##	trt5 / trt8	5.3267	1.2908	48	1	6.903	<.0001
##	trt5 / trt9	7.2075	1.7466	48	1	8.151	<.0001
##	trt7 / trt8	0.2726	0.0660	48	1	-5.364	0.0001
##	trt7 / trt9	0.3688	0.0894	48	1	-4.116	0.0035
##	trt8 / trt9	1.3531	0.3279	48	1	1.248	0.9129
##							

```
## P value adjustment: tukey method for comparing a family of 8 estimates
```

```
## Tests are performed on the log scale
```